



Quality Report

Sample 200042391

Report for Sample 200042391 created: May 5, 2026

Gene panel HCS_v2_hg38: 84 genes | 1,350 target regions | 284,897 bp

Number of samples: 8

Contents

1	Sample 200042391-0-S3	2
1.1	Dataset	2
1.2	Raw reads quality	2
1.3	Alignment statistics	5
1.4	Soft clipping	6
1.5	Mapping statistics for HCS_v2_hg38 gene panel	8
1.6	Target region coverage for HCS_v2_hg38 gene panel	9

Sample 200042391-0-S3

1.1 Dataset

For sample 200042391-0-S3 2,530,000 reads were sequenced, of these 2,511,050 were mapped corresponding to 99.25% of all reads.

1.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 1.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 1.2 shows the distribution of average base call quality, while Figure 1.3 shows the distribution of read lengths.

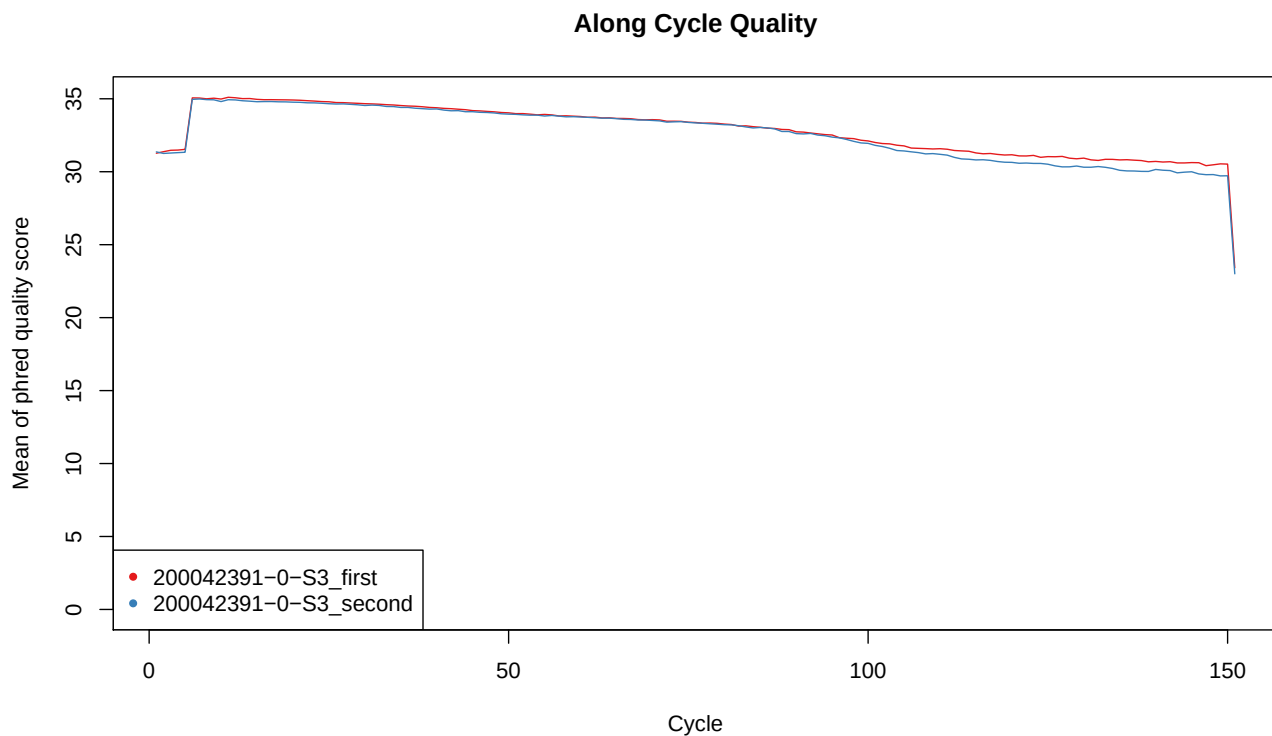


Figure 1.1: Mean base call quality by cycle.

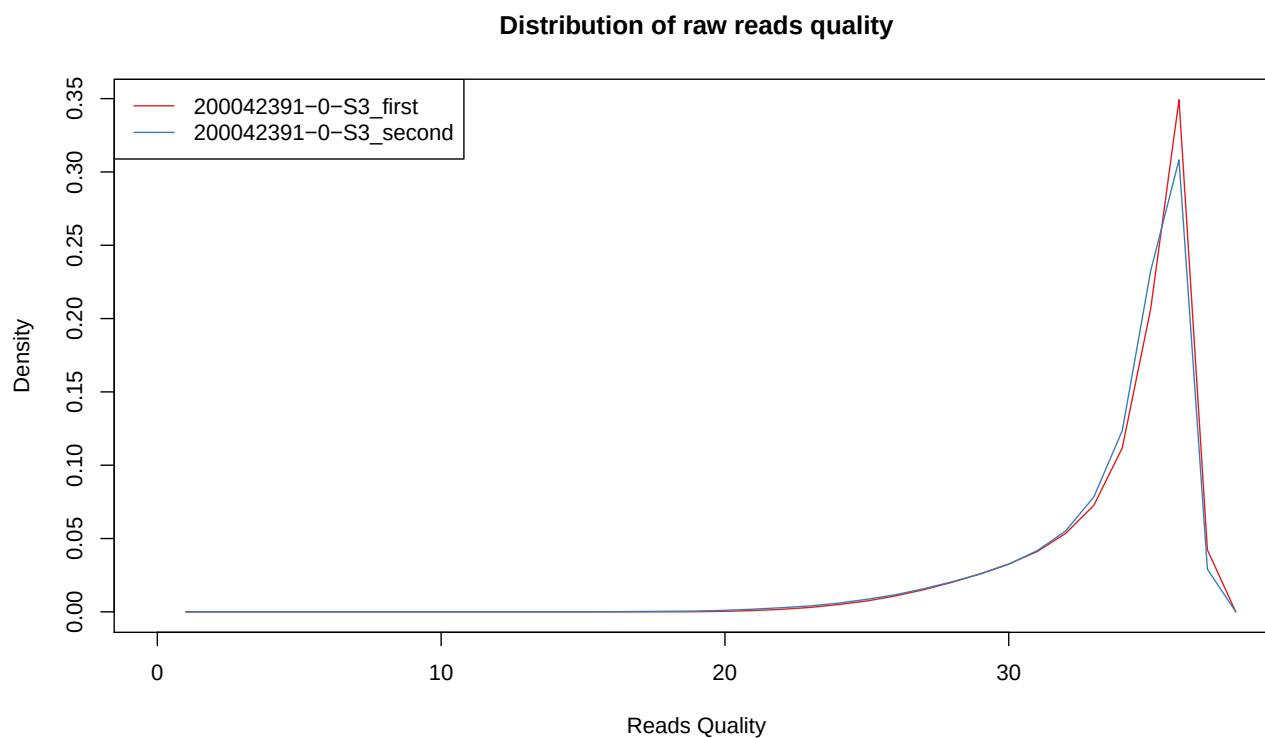


Figure 1.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

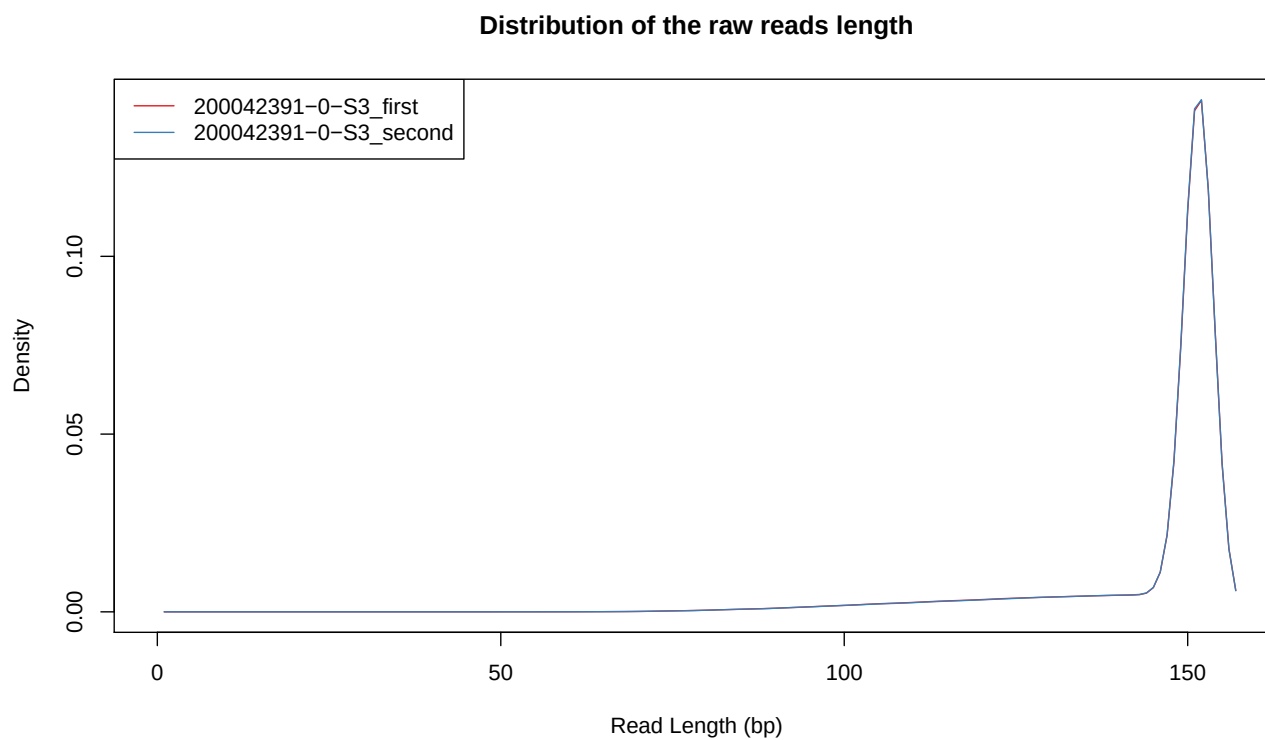


Figure 1.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

1.3 Alignment statistics

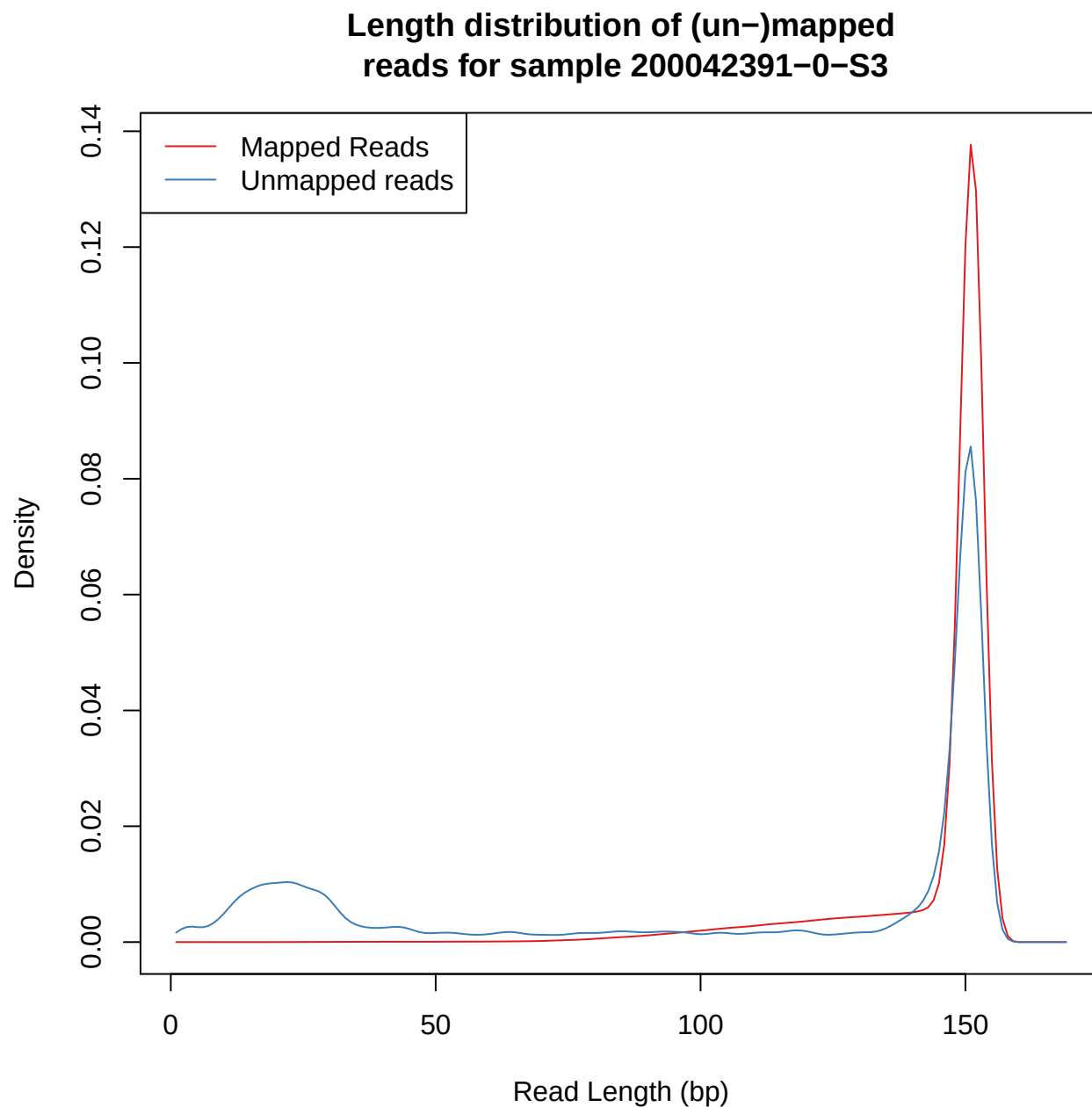


Figure 1.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

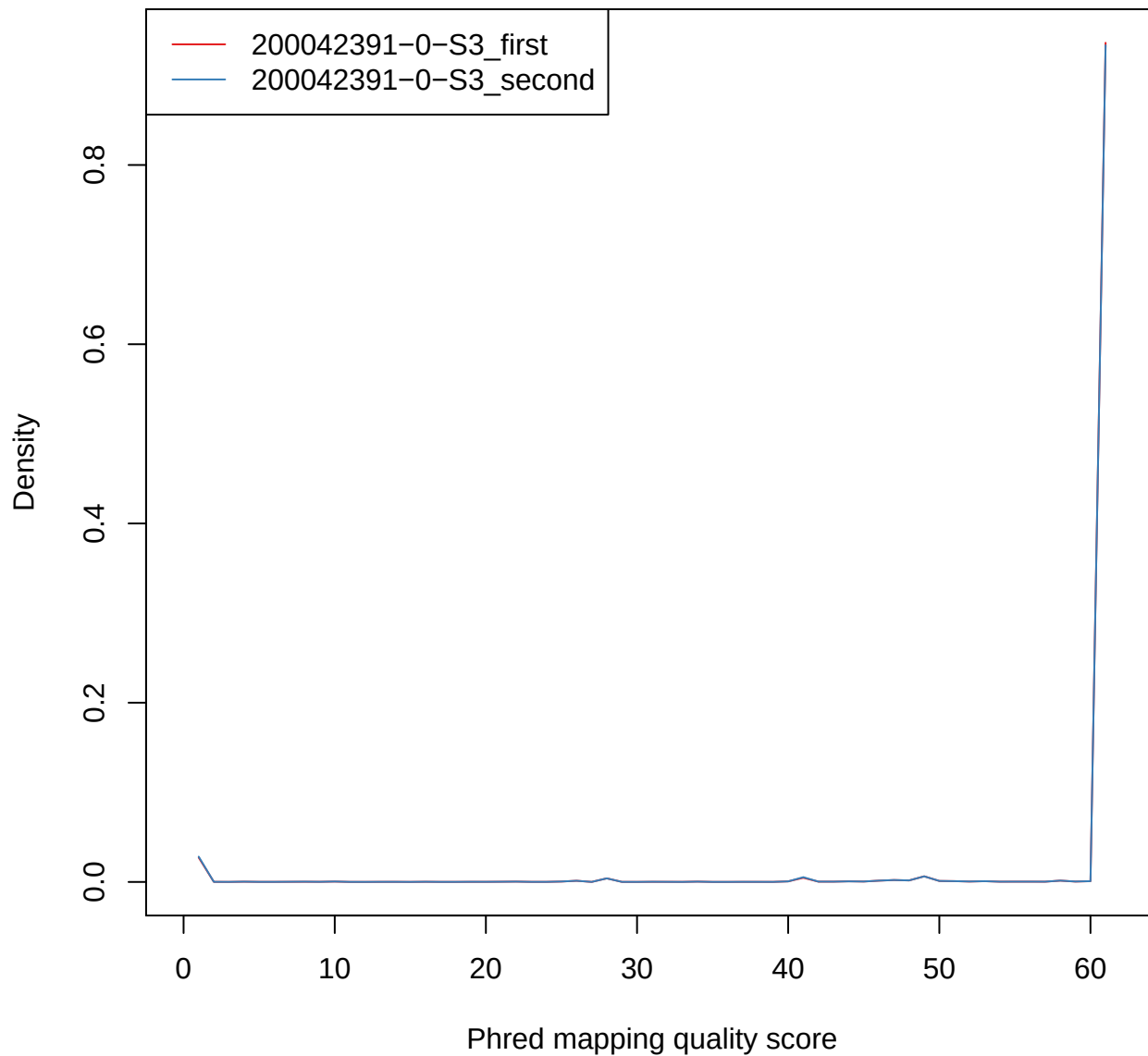


Figure 1.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

1.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042391-0-S3 (2,511,050 aligned reads)**

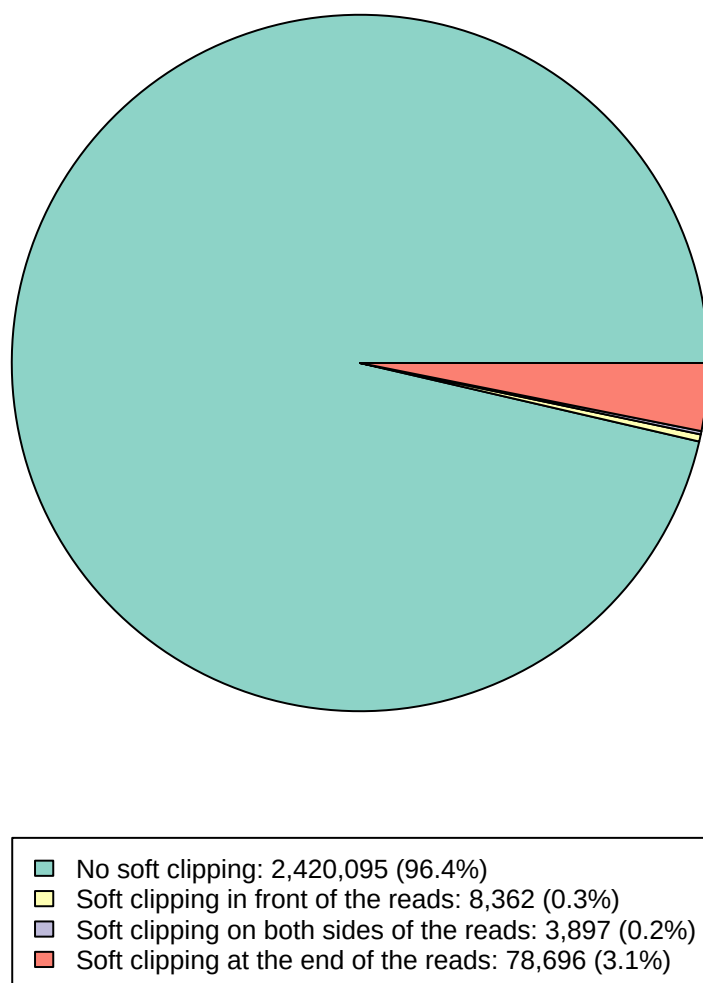


Figure 1.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042391-0-S3

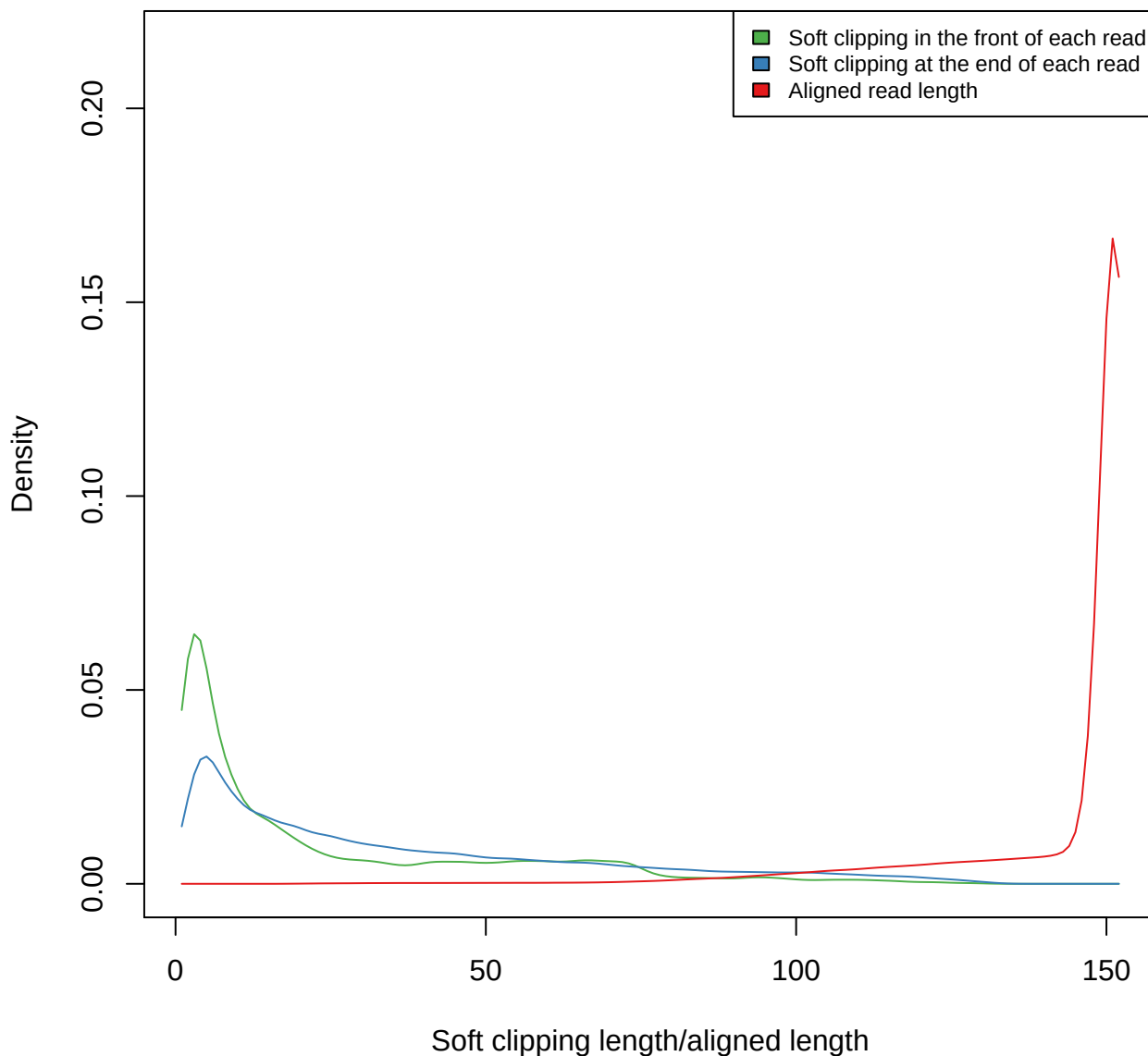


Figure 1.7: Density plots of soft clipping and aligned read length.

1.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 1.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042391-0-S3	74.19%	11.62%	7.48%	6.70%

Table 1.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042391-0-S3	4.65%	51,972	2,614	141	4	0	0	0

1.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 1.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042391-0-S3	1

Table 1.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	96	MSH2	NM_000251.3	8	c.1277-3297_1277-3202	14

Target region coverage distribution for sample 200042391-0-S3

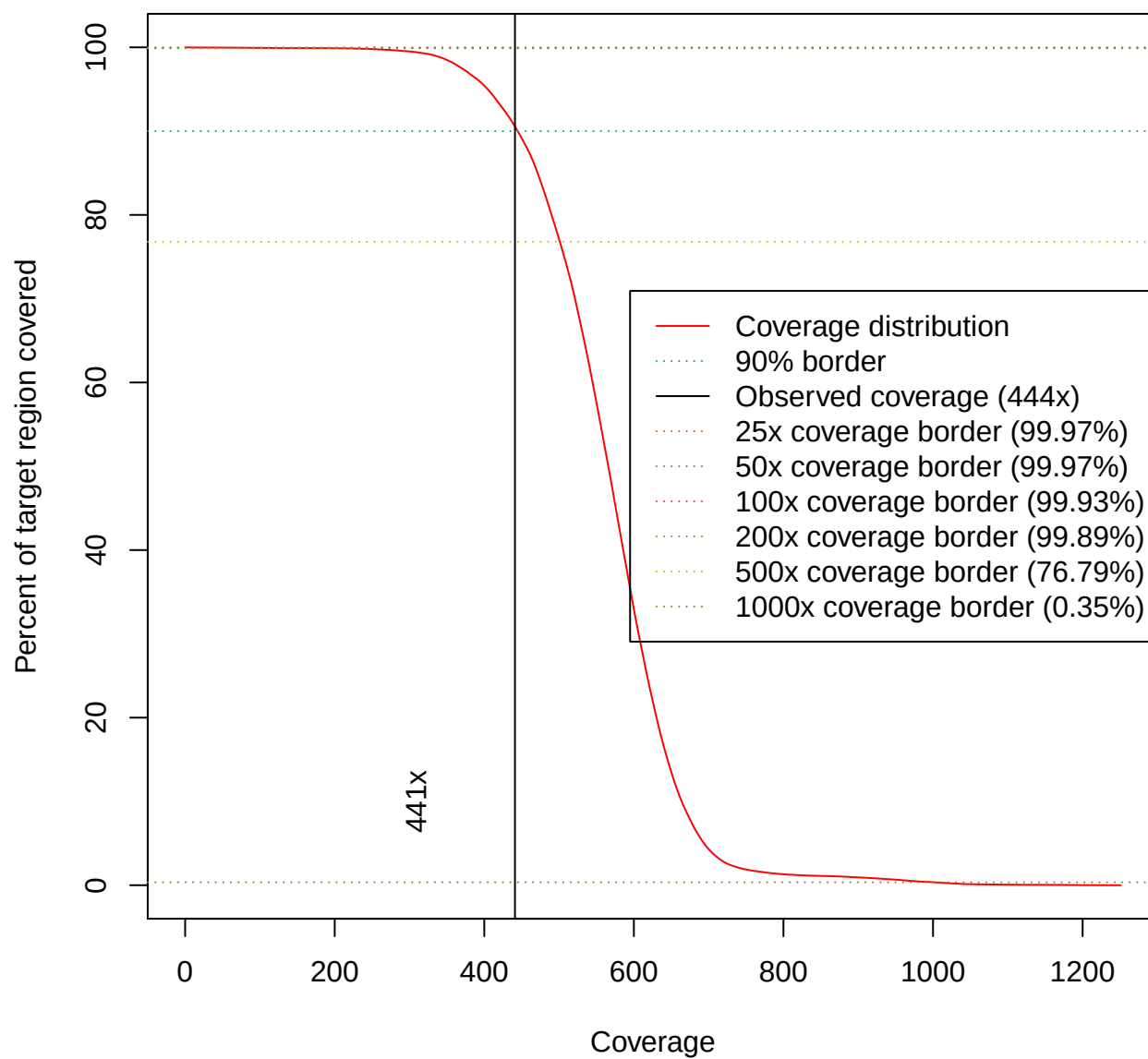


Figure 1.8: Target region coverage.

Table 1.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042391-0-S3	99.97%	99.97%	99.93%	99.89%	76.79%	0.35%	444×	0.08%

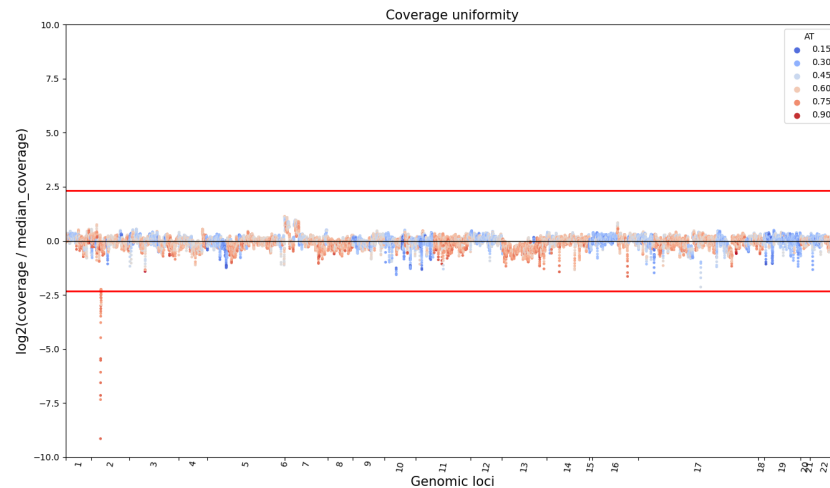


Figure 1.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.